

TARUN L. SANKAR

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EDUCATION

The University of Texas at Dallas - Dallas, TX | Expected May 2027

B.S. in Computer Engineering

GPA: 3.7/4.0 | Collegium V Honors | Dean's List | National Merit Scholar

EXPERIENCE

NorthMark Compute & Cloud - Dallas, TX | Jun 2026 - Aug 2026

Software Engineering Intern, Network Automation

- Replaced reliance on manual log checks and outage reports with a production Python/OpenObserve alerting service, giving 10+ network engineers early visibility into Infracore and zero-touch provisioning (ZTP) issues; delivered 1,000+ Slack alerts and surfaced 200+ ZTP errors over five weeks.
- Designed a command-driven Infracore snapshot-comparison harness that flags changes from baseline state and posts configuration diffs to Slack.
- Developed a tool-calling diagnostic agent used in network investigations, integrating Jira, Slack, Infracore, and ZTP systems; deployed the service through Harness and Argo-based CI/CD.

University of Texas at Dallas - Dallas, TX | Aug 2024 - Present

Undergraduate Researcher

- Developed a multithreaded POSIX C pipeline on a Raspberry Pi to compute S4 scintillation indices every five minutes from GNSS measurements sampled at up to 20 Hz, while receiver logging continued.
- Integrated Azure Blob Storage and Dash for storing and visualizing processed indices; one observed day yielded 4.5 MB of indices from 2.4 GB of raw measurements, reducing data volume by >99.8%.

MyIntent.io - Remote | Aug 2024 - Nov 2024

Software Engineering Intern, Full Stack Development

- Built the document-upload UI for a Next.js/NestJS RFP automation tool that generates business documents from structured customer inputs.
- Implemented document storage and retrieval for retrieval-augmented generation (RAG), enabling the application to use uploaded documents as context for business-document generation.

PROJECTS

Orune - Generative Educational Animation | Aug 2026

useorune.com

- Built and publicly launched a web app that turns text prompts into narrated, downloadable Manim videos, using React/TypeScript, Express, and Python.
- Implemented frame annotations and model-assisted review of sampled frames, enabling video revisions while preserving immutable render history.
- Queued rendering jobs through Cloud Tasks for execution in isolated E2B workers, stored artifacts privately in Google Cloud Storage, and implemented usage credits with Stripe billing.

Cinder Control - TAMUHack 2026 | Jan 2026

2nd Place, NorthMark HPC Sponsor Track; Best Solo Project

- Developed a wildfire simulation prototype solo in 36 hours using Next.js and a Python/FastAPI backend.
- Modeled fire spread with NumPy cellular automata using fuel, wind, and slope; visualized scenarios in Mapbox and flagged nearby infrastructure using OpenStreetMap data.
- Enabled voice and text scenario exploration through OpenAI and ElevenLabs, incorporating NOAA/NWS weather data into the simulation interface.

RESEARCH PRESENTATION

First Author, Poster - "Real-Time Scintillation Monitoring using Low-Cost GNSS-Based Monitors."

2025 Joint CEDAR/GEM Workshop, Des Moines, IA.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C, C++, Java, SQL, Bash

Web & Backend: React, Next.js, Node.js, Express, NestJS, FastAPI, PostgreSQL

Cloud & DevOps: Google Cloud, Azure, Docker, Kubernetes, Terraform, Harness, Argo, OpenObserve

AI & LLMs: OpenAI API, RAG, LLM tool calling, Model Context Protocol (MCP) tool development

AI-Assisted Development: Claude, Codex, GitHub Copilot

Scientific & Media: NumPy, Dash, Manim, FFmpeg